

## Assignment: Time Response

The textbook exercises, designated with the word “Nise”, are from the “Problems” section at the end of Chapter 4 in *Control Systems Engineering* by Norman Nise, 7th edition.

This assignment uses exercises from the textbook. The additional instructions are to clarify and/or expand each exercise.

### Nise 4.2 and 4.3: Step response of first-order system

For the MATLAB plots, add annotation (by hand or using the computer) to indicate the *time constant* ( $\tau$ ) of the system.

### Nise 4.8: Pole-zero maps and step responses

- Generate the pole-zero plots by hand; a simple sketch of the s-plane is sufficient.
  - The MATLAB function `roots()` finds roots of polynomials (useful if you want to avoid the quadratic formula).
  - Optional: `pzmap()` checks your work directly from the transfer function.
- Use the MATLAB `step()` command to plot the step response. The goal is to build intuition connecting pole-zero locations to time-response character.
- The *general forms* referred to in the problem statement are:

$$c(t) = K[1 - e^{-at}]$$

$$c(t) = K[1 - Ae^{-\sigma_1 t} - Be^{-\sigma_2 t}]$$

$$c(t) = K[1 - Ae^{-\sigma_d t} \cos(\omega_d t - \phi)]$$

$$c(t) = K[1 - A \cos(\omega_1 t - \phi)]$$

$$c(t) = K[1 - Ae^{-\sigma_1 t} - Bte^{-\sigma_1 t}]$$

Without computing the full inverse Laplace transform, identify numeric values for  $a$ ,  $\sigma_1$ ,  $\sigma_2$ ,  $\sigma_d$ ,  $\omega_d$ ,  $\omega_1$  from the appropriate general form. You do not need to find the coefficients  $K$ ,  $A$ ,  $B$  (they are both tedious to compute by hand and not necessary for understanding the time-response characteristics).

### Nise 4.20 and 4.21: Response of second-order system

Parts (a) and (b) only.

### Nise 4.20: Computing performance metrics

#### Rise Time Approximation: Second-Order Underdamped System

For a second-order underdamped system with transfer function

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2},$$

“a precise analytical relationship between rise time and damping ratio,  $\zeta$ , cannot be found” [1, Sec. 4.6]. Consequently, we need a method to approximate the rise time.

The Nise textbook offers both a polynomial approximation and a graphical method for estimating the rise time. The polynomial approximation from the textbook (footnote 5 to Section 4.6):

$$\omega_n T_r \approx 1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1 \quad (0 < \zeta < 0.9, \text{ error} < 0.5\%)$$

Alternatively, Figure 4.16 in the textbook can be used as a graphical lookup table for estimating the rise time.

An alternative coarse (but useful) approximation from Franklin, Powell, and Emami-Naeini [2, Sec. 3.4], derived for  $\zeta \approx 0.5$ , is

$$T_r \approx \frac{1.8}{\omega_n}$$

For design work, where the relative rise-time performance between different designs is more important than the exact value, this coarse approximation can be quite useful.

For the purposes of class, the coarse approximation is sufficient, but students are free to use the more precise polynomial approximation if desired. Do remember that these approximations only hold for underdamped second-order systems in the form shown above.

### **Nise 4.32: Step response from pole-zero description**

Check your answers by using MATLAB to plot the step response for your answers. Verify that the generated plots agree with the graphs in the exercise.

### **References**

- [1] N. S. Nise, *Control Systems Engineering*, 7th ed. Hoboken, NJ: Wiley, 2015.
- [2] G. F. Franklin, J. D. Powell, and A. Emami-Naeini, *Feedback Control of Dynamic Systems*, 6th ed. Upper Saddle River, NJ: Pearson, 2010.